Ahmad Jaka 6.111 P-Set 3:

Problem 1)

t_{cd} = 35 ps

t_{pd} = 185 ps

Critical Path = $m_0 - m_2 - m_4 - m_5$

Problem 2)

```
`default_nettype none

module pulser(
    input wire trigger_in, clk_in,
    output logic pulse_out
);
    logic prev_state;
    always_ff @(posedge clk_in) begin
        if(prev_state) begin
            pulse_out <= 0;
            prev_state <=  trigger_in;
        end else begin
            if(trigger_in) begin
                pulse_out <= 1;
                prev_state <= 1;
            end
        end
    end
endmodule

`default_nettype wire
```

Problem 3)

3.1) 8'b1010_0101

1) There are 3 0→1 transitions & 3 1→0 transitions

2) 8'b1110_1111

3) 8'b0001_0001

4) option 1 has fewer

sent: 9'b1_1110_1111

3.1 cont) 8'b1111_1111

1) There are 0 0→1 & 1→0 transitions

2) 8'b0000_0001

3) 8'b1111_1111

4) option 2 has fewer

sent: 9'b0_1111_1111

3.2)

```
`timescale 1ns / 1ps
//////////////////////////////////////////////////////////////////////////////////
// Company:
// Engineer:
//
// Create Date: 09/19/2021 03:17:41 PM
// Design Name:
// Module Name: PSET3
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
//////////////////////////////////////////////////////////////////////////////////

`default_nettype none

module tallier(
    input wire [7:0] byte_in,
    output logic [3:0] tally_out
    );
    assign tally_out = byte_in[7] + byte_in[6] + byte_in[5] + byte_in[4] + byte_in[3] + byte_in[2]
+ byte_in[1] + byte_in[0];
endmodule

`default_nettype wire
```